Journal of Building Engineering 111 (2025) 113090

Contents lists available at ScienceDirect

Journal of Building Engineering

journal homepage: www.elsevier.com/locate/jobe

Gene-modified acoustic metamaterials for improving the low-frequency broadband noise reduction performance of sound barriers for transportation buildings

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A R T I C L E I N F O

Keywords:
Acoustic metamaterials
Gene-modified
Sound barriers
Low-frequency broadband
Noise reduction

A B S T R A C T

To improve the noise reduction capability of sound barriers for existing transportation buildings, a gene-modified acoustic metamaterial is proposed for noise reduction into the sound barriers. Simulation results show that the sound absorption coefficient of metamaterials sound barrier (MSB) is more than 0.85 in the range of 800Hz-8kHz, experiencing significant improvement across both low and medium acoustic frequency ranges compared with the existing sound barriers., wherein it is found that the impedance-matching effect is the key to achieving near-perfect sound absorption. The simulation results of sound absorption coefficient and sound transmission loss match well with experiments, where it is found that the sound insulation performance can reach higher than 30.0 dB at frequencies above 200 Hz from the sound transmission loss curves. Experiments demonstrate that the installation of metamaterial sound barriers (MSBs) can reduce ambient traffic noise by 7.3 dB. This research is expected to bring new technical support for reducing urban traffic noise and improving the indoor living environment.

1. Introduction

With the rapid development of modern transportation, the lines inevitably cross environmentally sensitive areas, and residents along the transportation routes are facing increasingly serious environmental noise interference [1]. Among them, the installation of sound barriers in noise-sensitive areas such as dense transportation lines and residential living areas has become a crucial noise reduction measure [2]. The core material of existing sound barriers is mostly a single porous sound-absorbing material, which can play a certain sound-absorbing effect in the middle and high frequency ranges. However, with the increase in demand for transportation networks and quiet living environments for residents, traditional sound barriers highlight significant limitations when dealing with multi-frequency noise: (1) Low-frequency failure problem: the absorption rate of existing single-layer porous materials for

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https://doi.org/10.1016/j.jobe.2025.113090
Received 8 March 2025; Received in revised form 25 May 2025; Accepted 2 June 2025
Available online 6 June 2025

low-frequency noise below 500Hz is generally less than 40 %, resulting in the persistent exceedance of low-frequency roar in sensitive areas [3]. (2) Limited peak sound absorption: the peak insertion loss of conventional sound barrier structures is mostly limited to 10 dB or less [4,5], which is difficult to meet the stringent standard of $\leq$55 dB for high-density residential areas.(3) Frequency band imbalance: the sound absorption coefficient of the existing materials can reach more than 0.8 in the middle and high frequency bands, but it plummets to less than 0.3 in the low frequency band of 800Hz [6,7].

Acoustic metamaterials [8–10] have attracted much attention in recent years as artificial acoustic materials that realize special physical properties by carefully designing their internal structures. Unlike traditional materials, acoustic metamaterials have excellent sound insulation or sound absorption properties due to their special structural design on the sub-wavelength scale, such as bandgap insulation properties [11], micro-perforated broadband sound absorption [12], low-frequency acoustic wave control [13], and negative acoustic refraction under the super-surface [14,15]. In the field of acoustic metamaterials, which produce sound insulation effects at sub-wavelength scales, Fang et al. verified the negative elastic modulus near the resonant frequency and its huge transmission loss through an array of Helmholtz resonant cavities, providing a new direction for the design of highly acoustic isolation devices [16]. Kim et al. proposed the design of acoustic windows based on Helmholtz resonance cavities, which achieves transmission loss in multiple frequency bands and maintains noise energy dispersion [17]. Melnikov designed a C-shaped resonant cavity, which successfully isolates mechanical noise and provides the necessary ventilation and heat dissipation, demonstrating the potential of this structure in noise control [18]. Cheng proposed a two-dimensional Mie resonant cylindrical structure, and its low-frequency resonance characteristics were revealed by eigenstate analysis [19]. Yang et al. embedded a folded pipe structure in an open flat plate and successfully reduced the sound insulation frequency [20]. Liu proposed an acoustic-ventilated sound-insulating cage based on the electromagnetic wave-like theory to achieve the special performance of high efficiency of sound insulation at sub-wavelength scales with wide frequency [21]. Merkel demonstrated that a Helmholtz resonance cavity through the coupling achieves perfect absorption and pointed out that the mechanism applies to all fluctuating systems [22]. Wu exploited the weak coupling effect between the near-neighbor resonance cavities to achieve efficient acoustic wave absorption at low frequencies while maintaining the ventilation function [23]. Lee proposed an ultra-sparse absorptive material, which, by optimizing its volumetric ratio, ensured a lightweight design of the structure while guaranteeing high noise control [24]. Su et al. designed a structure with an inversely placed resonance cavity that exhibits almost perfect absorption properties near the resonance frequency [25]. Kumar et al. matched the impedance of the material to the air by opening multiple neck tubes on the resonance cavity of the acoustic metamaterial. It was found that the metamaterial achieved greater than 96 % absorption at 1000 Hz [26,27]. Du et al. designed a double-layer labyrinth-type acoustic metamaterial to achieve different acoustic absorption properties in both directions through asymmetric loss [28]. Ma et al. successfully realized narrow-band acoustic wave isolation through the coupling of thin-film-type metamaterials with open-aperture structures, which provided a new design idea for acoustic insulation [29]. Zhang designed a folded pipe structure that effectively realized narrowband acoustic wave blocking near 5900 Hz by using the interference effect caused by phase delay [30]. Ghaffarivardavagh successfully suppressed low-frequency noise by designing a spiral pipe coupled with a central aperture structure using the interference phase cancellation effect of this acoustic metamaterial [31]. Wang et al. successfully suppressed low-frequency noise by designing an acoustic metamaterial film and introduced the air-hole acoustic field interference, which enhanced the acoustic wave-blocking effect and achieved a transmission loss of 20 dB at 430 Hz [32]. Sun et al. extended the acoustic bandwidth by integrating two interfering phase-elimination modes, further optimizing the noise reduction performance of the acoustic metamaterial barrier [33,34]. Taking advantage of the miniaturization and lightweight of acoustic metamaterials in sound absorption of rotating equipment, Bai et al. proposed an acoustic metamaterial structure composed of multidirectionally distributed units (MOAMi) [35], and studied its omnidirectional acoustic wave absorption characteristics. The results show that the structure has an absorption coefficient of more than 0.8 over a wide range of incidence angles and a tunable frequency range, and experimentally verified its efficient noise reduction capability. For the vibro-acoustic problems of circular beams widely used in aerospace and other fields, Deng and Gao [36] proposed a curved ABH design suitable for mounting on the inner wall of cylindrical shells, and utilized the non-negative sound intensity to identify the active acoustic field, which provides a new solution for vibro-noise control of complex structures. Firoozy's team proposed a quasi-periodic energy harvesting system based on magnetically levitated metamaterials [37], which realizes broadband energy capture through a time-lag modulation strategy. The metamaterial system utilizes nonlinear vibration to establish an efficient energy harvesting band in the non-resonant frequency domain, breaking through the limitations of the traditional resonance mechanism and significantly expanding the effective bandwidth of vibration energy harvesting. Ghoudjani's team breaks through the limitations of traditional acoustic materials and proposes a three-dimensional chiral honeycomb membrane metamaterial based on the acoustic circular dichroism negative mass effect and local resonance mechanism [38]. And the structural parameters were optimized by genetic algorithm to achieve a 35 % increase in average sound transmission loss in the low frequency band below 850Hz. Gao prepared underwater metamaterials with different aluminum foam scatterers, and improved sound absorption performance was achieved by manipulating underwater sound propagation paths and wave modes [39]. On this basis, Gao et al. proposed a composite metamaterial that can achieve sound absorption and electromagnetic wave absorption, and combined with an optimization algorithm to achieve multi-objective geometry optimization [40]. Both the optimal design methodology and experimental methodology of metamaterials proposed in this study have potential guiding value in the development of metamaterials.

With the growing demand for practical applications of acoustic metamaterials and the limitations of application scenarios, more efficient acoustic metamaterial design methods have received much attention. Ghorbany used a chaotic particle swarm algorithm (MOPSO) to enhance the pneumatic suspension handling system and ride quality based on a vehicle thermodynamic analysis model [41]. Zhang et al. proposed a ventilated noise-reducing metamaterial structure combined with an optimization algorithm for reverse design, which greatly improved the engineering design efficiency and ventilation and noise reduction performance [42]. Chang et al. developed a tunable acoustic metamaterial by combining genetic algorithm (GA) and neural network (NN) models. This method considers sensitivity analysis and improves the local optimum dilemma of particle swarm algorithm (PSO) to some extent, and realizes the bandgap-controllable wideband acoustic isolation design [43]. Wu et al. prepared a joint simulation program of MATLAB With COMSOL using GA and finite element method (FEM). Based on this method, a Helmholtz resonant acoustic metamaterial was designed, and both simulation and experimental results verified its low-frequency acoustic isolation characteristics. This study also further demonstrated the good applicability of GA in acoustic metamaterial design [44].

The above research has largely promoted the development of acoustic metamaterials theory and design methods, but there are still many deficiencies in low-frequency sound absorption and broadband noise reduction. In addition, acoustic barriers, as important noise reduction equipment in the fields of rail transportation, highway transportation, and building sound insulation, are also in urgent need of upgrading. How to combine the development of acoustic metamaterials with the efficient and optimized design and modular engineering application of new sound barriers is an important research topic.

Based on the above analysis, this study developed a metamaterial sound barrier (MSB) based on acoustic metamaterials using genetic optimization algorithm and agent model to improve the noise reduction genes of the sound barrier. Compared with the existing noise barriers, the MSB has significantly improved sound absorption performance in the low and medium frequency range of 200~1000 Hz, and the sound insulation performance can reach more than 30.0 dB at frequencies above 200 Hz. The research combines both simulations and experiments to ensure the modular application and engineering noise reduction effect of the metamaterial sound barriers (MSBs).

The research process of the paper is as follows. In Section 2, the intelligent design method of metamaterial sound barrier and the optimal configuration of acoustic metamaterial are introduced. In Section 3, the noise reduction mechanism of the acoustic metamaterial module is simulated and analyzed. In Section 4, the acoustic absorption and sound insulation performance of MSB are simulated and analyzed. In Section 5, the sound absorption coefficient and sound transmission loss of MSB are tested by reverberation and acoustic isolation room tests. In Section 6, the noise reduction indexes before and after the installation of MSBs on highways are tested and compared.

2. Methodology and modeling

A genetic algorithm program was prepared (code is shown in the Supplementary file) and COMSOL Multiphysics 6.0 was controlled through the COMSOL With MATLAB interface for acoustic metamaterials creation, simulation, and optimization. Fig. 1 shows the flow of the genetic algorithm as shown. First, the required acoustic metamaterial basic module is constructed. W is the width of the basic module, and H is the thickness. The top and bottom layers of the basic module are rock wool, and the middle layers are glass wool, composite structure, and glass wool from left to right, respectively. Where a is the cavity width of the composite structure and b is the thickness. The composite structure consists of an insert plate with a certain slope gradient and a foam particle filler, respectively. Where k is the slope of the inserts; t is the thickness of each layer of the inserts; r is the radius of the circle at the center of the inserts; and φ *is* the fill rate of the particulate filler. The original acoustic metamaterial basic module is the first parent generation. The initial acoustic metamaterial basic module is the first parent generation, where a, b, k, t, r, φ are the variable genomes, which are genetically evolved in continuous iteration. W and H are the constraint genomes, which can be constrained according to our actual needs and do not participate in the iterative evolution.

The acoustic equivalent circuit model for each part of the structure is shown in Fig. 2(a) where the plane wave incident pressure field P is the acoustic power source. The equivalent impedances of the first and third layers of porous material are Za and Zc, respectively. The equivalent impedance Zb of the second composite layer consists of three parts connected in parallel. These three parts are the impedances Zb1 and Zb3 provided by the porous sponges at the left and right ends, and the impedance Zb2 provided by the central cavity and the particle filler, respectively. Fig. 2(b) shows the equivalent impedance model of the acoustic metamaterial structure, and the acoustic equivalent circuits of the three-layer structure are combined by means of series connection. Through the above logical relationship, the final equivalent circuit analysis model can be derived as shown in Fig. 2(c).

Based on the acoustic equivalent circuit model, the equivalent acoustic impedance Z_{all} of this acoustic metamaterial structure can be calculated as Eq. (2).

$$Z_{\text{all}} = Z_a + Z_c + \left({Z_{b1}}^{-1} + {Z_{b2}}^{-1} + {Z_{b3}}^{-1}\right)^{-1} \tag{2}$$

From this, the sound absorption coefficient SAC of the structure can be further calculated as in equation (3).

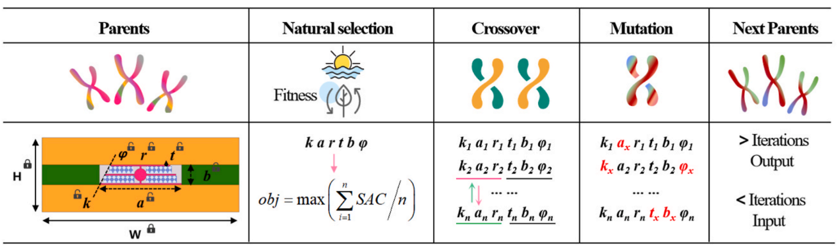

Fig. 1. Genetic algorithm flow for acoustic metamaterials.

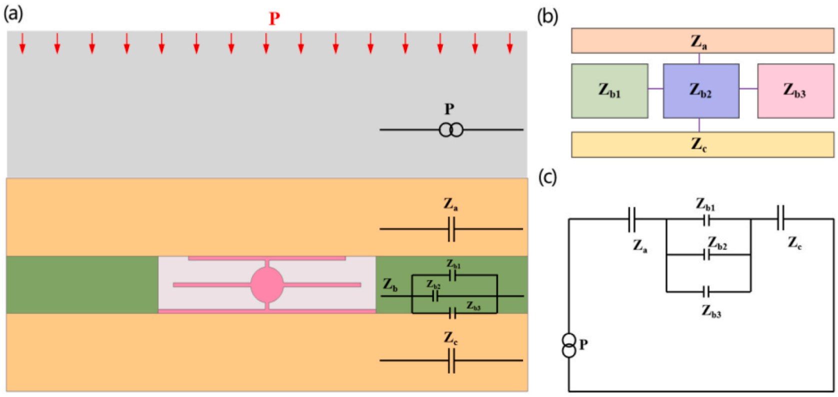

Fig. 2. Theoretical modeling of acoustic equivalent circuits.

$$SAC = 1 - \left|\frac{Z_{\text{all}} - \rho_0 c_0}{Z_{\text{all}} + \rho_0 c_0}\right| \tag{3}$$

Where, ρ_0 is the density of air and c_0 is the speed of sound in air. All the above parameters can be calculated by COMSOL Multiphysics simulation.

Combined with the measured data of related urban traffic noise, the maximum sound absorption coefficient in the range of 100Hz–2000Hz is chosen as the adaptation function. This is the key to the natural selection of variable genomes. After the selection, the genome will be programmed to undergo gene crossover and mutation [45,46] to obtain more gene combinations. Finally, the new genome undergoes natural selection again and continues to retain the advantageous genes. This cyclic process stops until the requirement of the number of iterations is satisfied. The final output of advantageous genes can form the optimal configuration of the metamaterial structure.

The fitness function, i.e., the objective function, is shown in equation (4).

$$fit = \max\left(\sum_{i=1}^{m} SAC \Big/ m\right) \tag{4}$$

Where *fit* is the fitness function. *m* is the number of frequencies to be weighted. *SAC* is the sound absorption coefficient at a certain frequency.

Fig. 3 shows the evolutionary process of acoustic metamaterial evolution. The fitness curve in Fig. 3(a) shows that the fitness of acoustic metamaterial genes increases with the increase in the number of evolutions. When the number of iterations reaches 60, the fitness curve gradually stabilizes and reaches 0.97 at the 80th iteration. The genomic structures of acoustic metamaterials and their corresponding average sound absorption coefficients are shown in Fig. 3(b) for the four stages of the evolution process, namely, A, B, C, and D. The structural configuration corresponding to the D stage, which is the optimal one in this research, is shown in Fig. 3(b). Fig. 4(a) shows the optimized acoustic metamaterial surface as well as a 3D model view of the metamaterial sound barrier (MSB) panel. The figure shows the details of the distribution of the two layers of rock wool, glass wool, built-in inserts, and foam particle filler. Details such as specific geometric parameters are shown in Table s1 of the Supplementary file. Fig. 4(b) shows the front, left, and top views of the metamaterial sound barrier panel. In this case, the built-in inserts and glass wool are arranged in a periodic arrangement pattern of A-B-A-B. Fig. 4(c) shows the rock wool material in actual production. Fig. 4(d) shows the internal structural details of MSB panels in actual fabrication, in which the built-in inserts and glass wool, etc., are consistent with the ideal design model. Fig. 4(e) shows the specimen of the fabricated built-in insert plate. Fig. 4(f) shows the built-in insert after filling with foam particles. Combined with the design requirements of the sound barrier, the actual engineering configuration of the metamaterial sound barrier MSB can be obtained

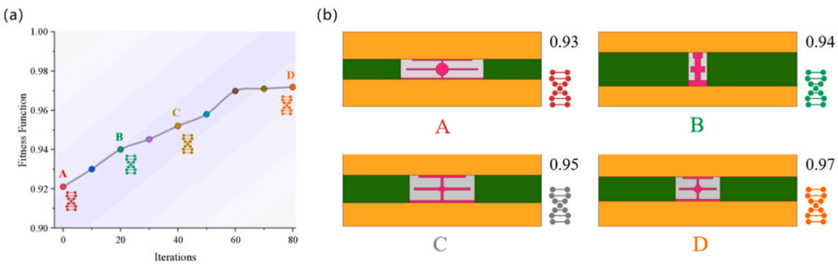

Fig. 3. Evolution of acoustic metamaterials (a) Adaptation curves (b) Comparison of genome structure at different stages.

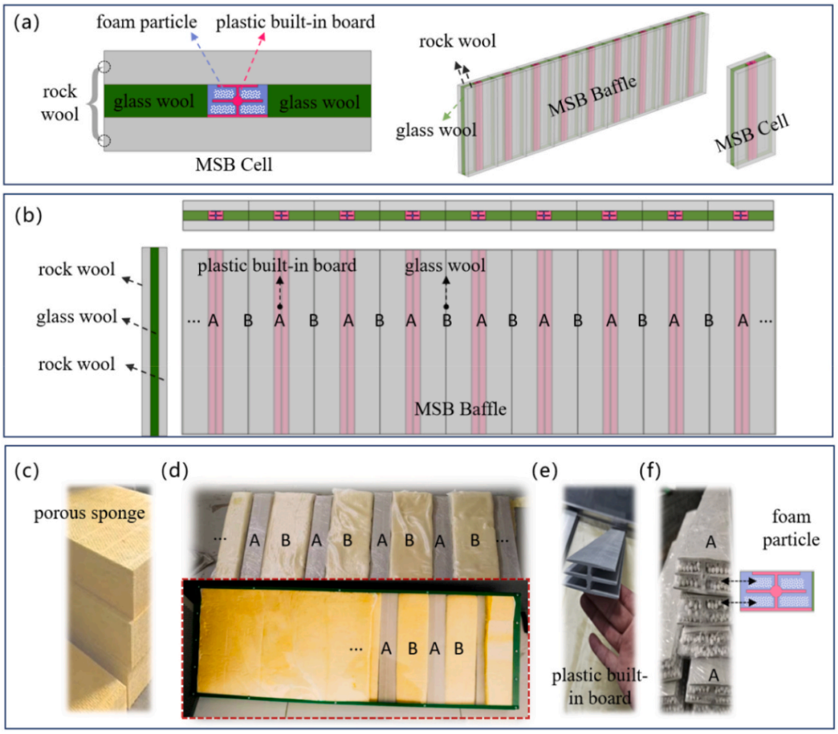

Fig. 4. MSB building block details. (a) Cell composition. (b) Three-dimensional view of the ideal model of MSB Bafflle. (c) Rock wool sample. (d) MSB Baffle sample. (e) Built-in insert. (f) Foam particles.

by assembling the four pieces of MSB Bafflle with the columns and bolts as shown in Fig. 5(a). Fig. 5(b) shows the field sample schematic of the column. Fig. 5(c) shows the ideal installation model of MSB. Fig. 5(d) shows the schematic diagram of the on-site installation of the Hubei highway, which is 150m long in total. It can be seen that in the field application of transportation engineering, MSBs usually have to be installed in a continuous arrangement to play a practical role in noise reduction.

3. Acoustic simulation

The optimized MSBs were subjected to further multiphysics field simulation through the acoustic-solid mechanics multiphysics field module of COMSOL Multiphysics software. In particular, the built-in insert is set as a solid field with material properties of ABS plastic; the acoustic parameters of rock wool, glass wool, and foam particles are selected as equivalent to the JCA model [47,48]. The modeled equivalent parameters of glass wool, rock wool, and foam particles used in the research were all provided after testing by Hunan Jiuwei Tongchuang Polymer Materials Co. Specific experimental parameters are shown in Table S2 in the Supplementary

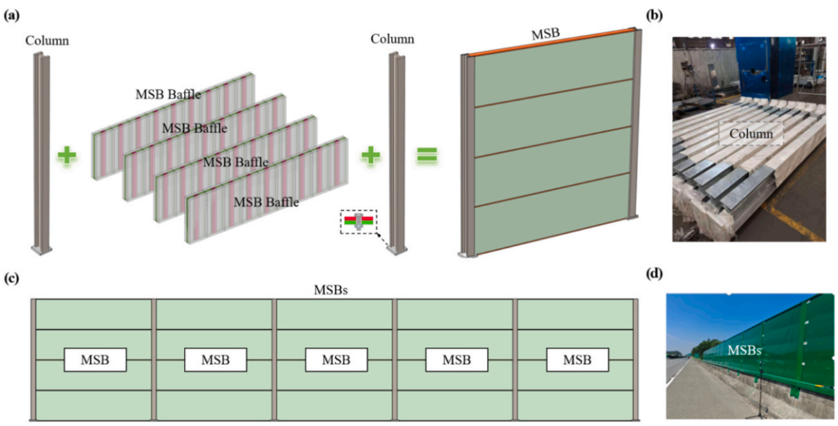

Fig. 5. (a) Components of the MSB. (b) Sample of the riser. (c) Ideal model of MSBs. (d) Field solid model of MSBs.

documents. The rest of the cavity was simulated with the air domain in the pressure acoustic domain. The finite element simulation was performed with a customized grid, and the minimum grid cell should be less than 1/6 of the corresponding wavelength. A customized grid was used in the actual simulation, and the minimum grid cell was set to be 1/10 of the maximum wavelength, which meets the requirement of setting the grid at a wavelength of less than 1/6. The radiation condition is set as plane wave incident. The amplitude of sound pressure in the incident region is 1 Pa. By extracting the incident acoustic pressure field and scattered acoustic pressure field at each frequency, acoustic indexes such as absorption coefficients can be calculated. Details of the simulation model and boundary condition settings in COMSOL Multiphysics can be found in Fig. S1 in the Supplementary documents.

Where the sound absorption coefficient is defined as in equation (5).

$$SAC = 1 - \left(\frac{P_s}{P_i}\right)^2 \tag{5}$$

Where P_s is the scattered sound pressure and P_i is the incident sound pressure.

The acoustic incidence, acoustic absorption, and acoustic reflection occurring in the MSB plate are schematically shown in Fig. 6(a). The simulation results for the unit cell, the 8-element cell, and the Floquet periodic boundary condition [49–51] set are compared in Fig. 6(b). The mathematical formal solution of the Floquet periodic boundary condition needs to be satisfied:

$$p(x+a, t+T) = e^{i(ka-\omega T)}p(x,t) \tag{6}$$

where x is a spatial quantity, t is a temporal quantity, a is the period in the spatial dimension and T is the period in the time dimension.

Due to the increase of cavity area, the acoustic absorption effect of the 8-member cell structure is slightly enhanced in the range of 2000 Hz–4000 Hz. The acoustic absorption effect of the 8-periodic cell structure is very close to that of the cell structure with a periodic boundary. In the range of 200Hz–8000Hz, the sound absorption coefficients of the three are more in line with each other. SAC at 500 Hz and 800 Hz is above 0.6 and 0.85 respectively. The above results demonstrate that the MSB panels have wide-frequency and high-efficiency acoustic absorption performance. Fig. 6(c) shows the scattered sound pressure fields at different frequencies of the MSB plate for 8 cells. The scattered sound pressure level at the incident end of the sound wave at different frequencies is smaller than that in the MSB region; the larger the SAC value is, the more obvious the difference is. This result indicates that the acoustic energy can be well absorbed by the MSB plate, which leads to a decrease in the sound pressure level in the incident region and an increase in the scattered sound pressure level of the MSB plate.

Fig. 7(a) compares the sound absorption coefficients of the MSB and the existing sound barrier (ESB) under the same parameter conditions. Compared with the ESB, the overall absorption coefficients of the MSB are greatly improved, especially in the low and medium frequency range of 200∼1000 Hz, where the SAC is significantly improved.

Fig. 7(b) extracts the sound pressure level field and sound power flow at 500 Hz and 2000 Hz for the ESB and MSB, respectively. Compared with ESBs, MSBs have a “siphoning effect” or “resonance effect”, where acoustic energy is more easily aggregated and the sound absorption performance is better; the markings in Fig. 7(b) are made using white circular areas. For example, at 500 Hz, a trapezoidal siphon effect occurs at the bottom of the acoustic metamaterial composite region, which interferes with the movement of acoustic energy in the original porous structure; while at 2000 Hz, multiple modes of resonance coupling effects occur in each region block due to the matching design of the acoustic metamaterial composite region with multiple sound-absorbing regions. The appearance of the siphoning and resonance effects enhances the acoustic wave aggregation and acoustic energy dissipation in the MSB structure, thus improving the acoustic absorption performance of the structure. In order to further investigate the sound absorption mechanism of the MSB, the real and imaginary parts of the normalized acoustic impedance are extracted for the MSB in Fig. 7(c). It can be seen that the real part of the MSB acoustic impedance gradually fluctuates around 1 as the frequency increases, while the imaginary part of the acoustic impedance gradually starts to fluctuate around 0. The real part of the MSB acoustic impedance is the same as the imaginary part of the MSB. When the coordinate value is closer to (1,0), its sound absorption coefficient is closer to perfect sound absorption. The circular correspondence between the real and imaginary parts of the MSB acoustic impedance is shown in Fig. 7(d). We

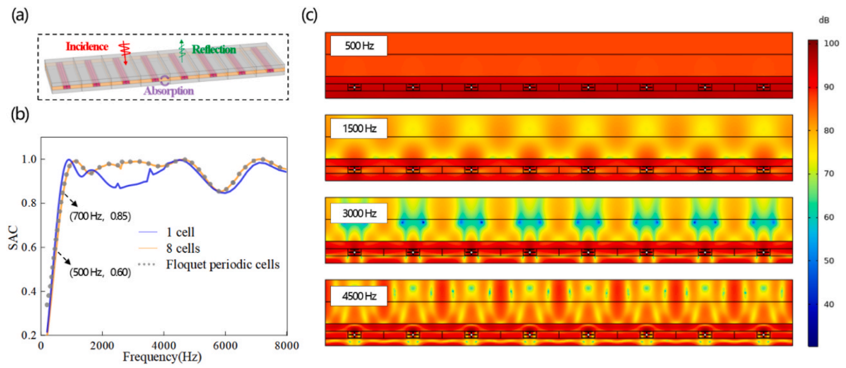

Fig. 6. Acoustic absorption simulation of MSB plate. (a) Schematic of sound absorption. (b) Sound absorption coefficient. (c) Scattered sound field at different frequencies.

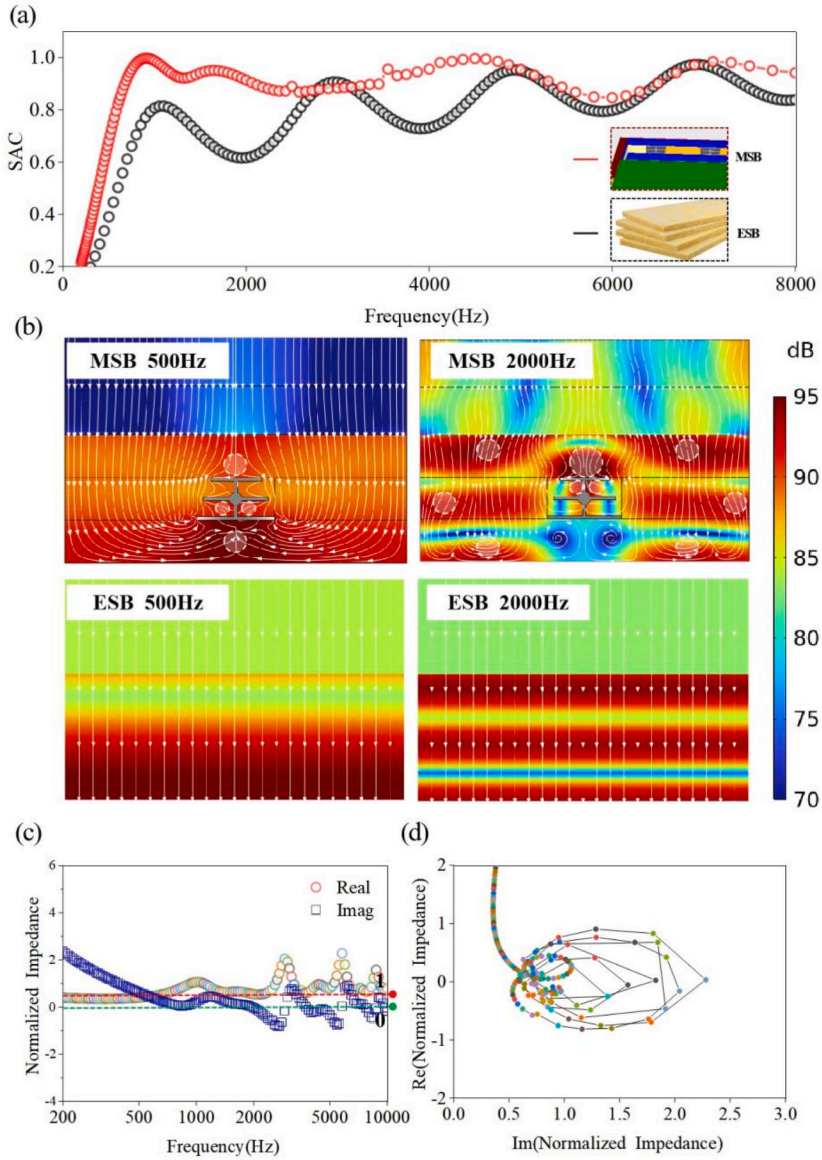

Fig. 7. Acoustic Simulation Results (a) Comparison of SAC curves of two structures (b) Comparison of sound pressure level cloud plots at different frequencies (c) Acoustic impedance real and imaginary parts of MSB. (d) Acoustic impedance loop diagram of MSB.

notice that both the real and imaginary parts of the acoustic impedance of the MSB surround around the coordinate (1,0), so the structure can achieve better sound absorption.

It can be found that impedance matching is the key factor in realizing perfect sound absorption. The above analysis results show that the impedance matching between the real part and the imaginary part will enable the acoustic energy to enter the MSB easily, which greatly improves the acoustic absorption effect of the model. The acoustic metamaterial evolution process using genetic algorithms is exactly the optimization process of structural impedance matching design. When the equivalent acoustic impedance of the material is closer to the characteristic impedance of the air, the smaller the reflection coefficient of the acoustic wave at the interface. Eventually the acoustic absorption coefficient of the metamaterial is enhanced in the target frequency band.

In practical noise reduction engineering applications in addition to sound-absorbing structures, we usually use sound-insulating structures for noise blocking as well. Therefore, in this section, the sound insulation characteristics of MSB structures will be evaluated by calculating the sound transmission loss (STL), which is defined as follows.

$$STL = 10\,\lg\left(\frac{W_i}{W_s}\right) \tag{7}$$

Where W_s is the transmitted sound power and W_i is the incident sound power.

Fig. 8(a) shows a schematic of the sound insulation modeling calculations (More details can be found in Fig. S2 in the supplementary documents). Fig. 8(b) shows the STL calculation results. It can be seen that the STL value at frequencies above 200 Hz is above 30.0 dB; the STL value at frequencies above 800 Hz is stable at above 50.0 dB. The above results show that the MSB can effectively block or dissipate the propagation of acoustic energy.

4. Indoor acoustics experiment - testing SAC and STL

To test the actual sound absorption and sound insulation of the simulation-designed MSB. In this section, the SAC and STL of the acoustic metamaterial inserts are examined using the sound isolation chamber and reverberation chamber test methods. Fig. 9(a) shows the acoustic generation chamber, which includes the reverberation sound source and the microphone. Fig. 9(b) shows the mounting position of the sound barrier for the acoustic isolation test, surrounded by an airtight encapsulation. Fig. 9(c) shows the sound receiving room, which is built in for receiving the transmitted sound waves. Measurement of the STL can be accomplished by testing all three of the above. Fig. 9(d) shows the mini reverberation chamber, which is used to test the sound absorption coefficient. The data acquisition device is an INV3062 collector, as shown in Fig. 9(e). This collector uses an independent 24-bit per channel converter, which allows parallel processing of multiple channels. In this case, the maximum sampling frequency is 51.2 kHz.The data connection cable in Fig. 9(f) is used to connect the acquisition instrument to the microphone. The sound pressure microphone used is a transducer type INV9202. The microphone has a frequency response range of 20 Hz to 20 kHz and a maximum output of 134 dB SPL. The sample required for the test was an insert plate of the MSB, and this experimental sample had the same geometrical parameters as the simulated structure shown in Fig. 8(a). In particular, the length is 2.0 m, the height is 0.5 m, and the thickness is 0.08 m.

Fig. 10(a) shows that the experimental calculations of SAC are in good agreement with the simulation results. After 400 Hz, the SAC of both of them reaches more than 0.6. Above 800 Hz, the sound absorption coefficients of both of them are stabilized at about 0.85. Due to the nonlinearity of the material, the sealing of the device, and other disturbing factors, there are some differences between the two at the peak value. The above results verify that the MSB has the effect of low-frequency broadband sound absorption. As shown in Fig. 10(b), the experimental calculation results of the STL are consistent with the simulation results. The STL values at frequencies above 200 Hz are all above 30 dB. At frequencies above 800 Hz, the experimentally measured STL values are all above 50 dB. The above results also verify that the MSB has the effect of low-frequency broadband sound insulation. Due to the sealing nature of the sound insulation test and the sound reflection characteristics of different material impedances, the sound insulation test may have a slight error. It is worth noting that the acoustic phenomenon of peak fluctuation of the STL curve occurs due to the nonlinear thermo-viscous effect of the porous material, which increases the overall damping effect of the structure. This error phenomenon is also consistent with the conclusions in many references [12,31,37,38]. In summary, it is shown that the MSB possesses the advantages of low-frequency broadband sound absorption and low-frequency broadband sound insulation, which can realize near-perfect noise reduction. In addition, the modularized design method of MSB is expected to be applied in the future to noise reduction projects in the fields of automotive noise, railroad noise, aviation noise, industrial noise, and construction secondary noise.

5. Engineering applications and field testing

To further realize the effect of MSB's engineering application. We installed a 1000m long test section of highway sound barrier in the Jingyi highway section in Hubei. The calculation of insertion loss in the field is evaluated by the test method of relevant specifications. The sound barrier insertion loss D_{IL} is given by the following equation:

$$D_{\mathrm{IL}} = \left(L_{\mathrm{ref,A}} - L_{\mathrm{ref,B}}\right) - \left(L_{\mathrm{r,A}} - L_{\mathrm{r,B}}\right) \tag{8}$$

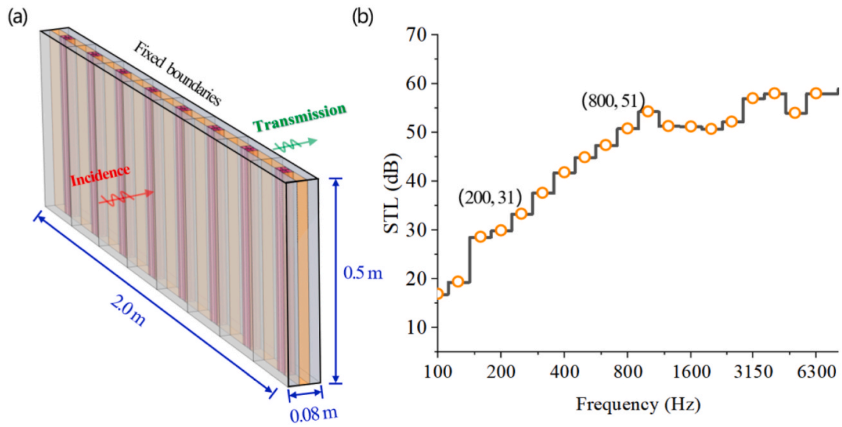

Fig. 8. Acoustic transmission loss. (a) Schematic of the simulation model. (b) STL.

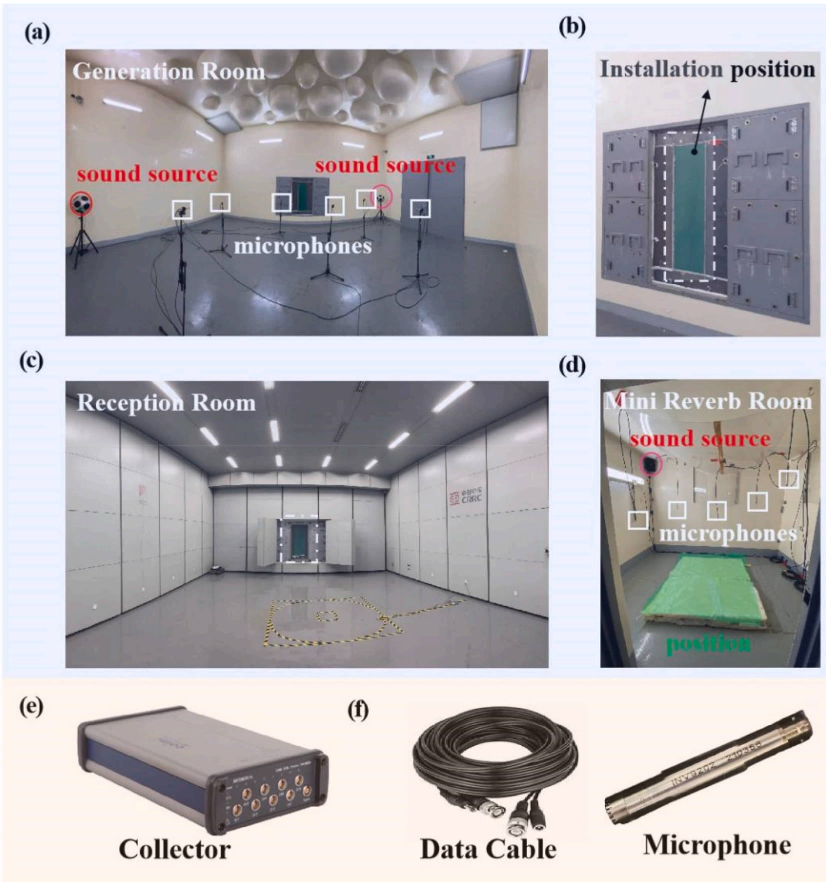

Fig. 9. Field experiment setup (a) Acoustic generation room. (b) MSB mounting position. (c) Acoustic reception room. (d) Reverberation room. (d) Collector. (d) Data cable and microphone.

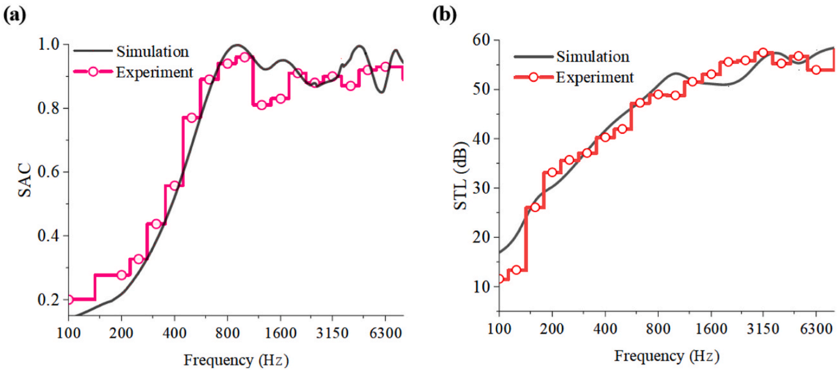

Fig. 10. Comparison of experiment and simulation. (a) SAC. (b) STL.

$L_{ref,B}$–Sound pressure level "before" the reference point is installed.
$L_{r,B}$– Sound pressure level "before" the installation of the receiving point.
$L_{ref,A}$–Sound pressure level "after" the reference point is installed.
$L_{r,A}$– Sound pressure level "after" installation at the receiving point.

Fig. 11(a) shows the test points before the installation of the MSB, where the microphone transducer is placed at S1 and R1 as the source and receiver points, respectively.11(b) shows the test points at the same location after the installation of the MSB, where S2 and

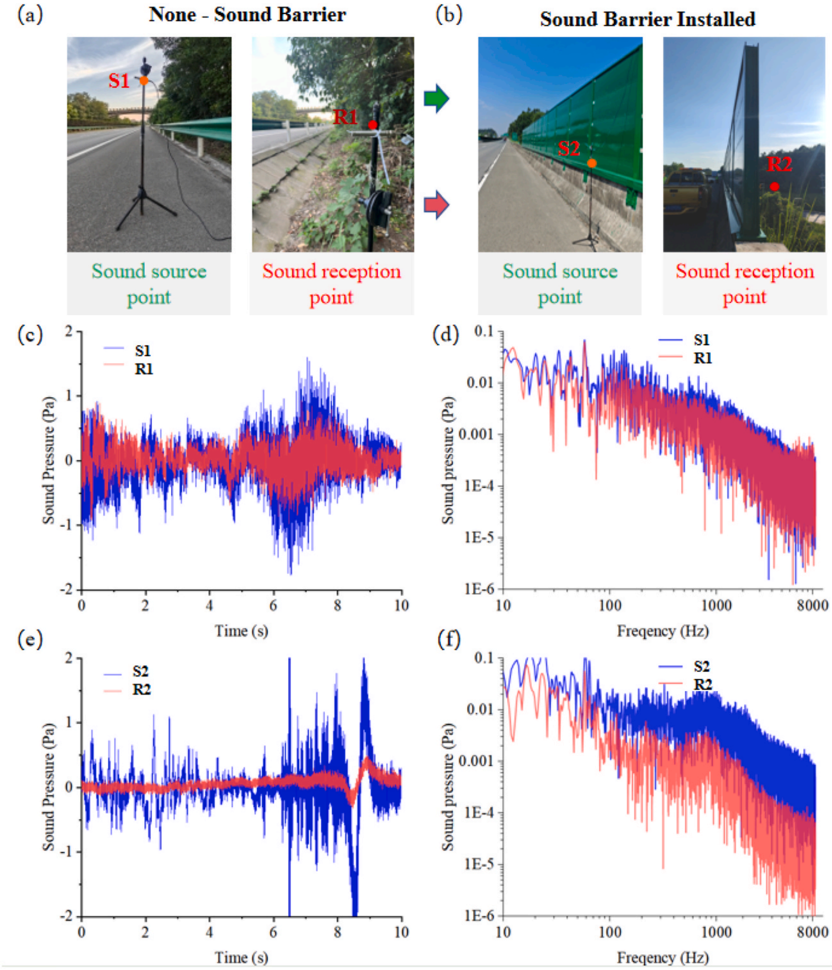

Fig. 11. Field test results (a) Measurement points without sound barriers. (b) Measurement points after installation of sound barriers. (c) Time domain of sound pressure for S1 and R1. (d) Frequency domain of sound pressure for S1 and R1. (e) Time domain of sound pressure for S2 and R2. (f) Sound pressure frequency domain of S2 and R2.

R2 are also the source and receiver points, respectively.

Fig. 11(c) and (d) show the comparison of the time and frequency domain results of the sound pressure at S1 and R1, respectively. It can be seen that the amplitude of the sound pressure in the time domain at S1 is slightly higher than that at R1, but the difference between the two is relatively small and is still in the same order of magnitude. In the frequency domain results can be seen, S1 point and R1 point of the spectrum of the difference is small, in the range of 10–100Hz amplitude is relatively close, in the range of 100–1000Hz there is a certain R1 point sound pressure is slightly lower than the S1 point. The above results show that the difference in noise amplitude between the sound source and sound receiving points is small before the installation of the sound barrier.

Fig. 11(c) and (d) show the comparison of the time domain and frequency domain results of the sound pressure at S2 and R2, respectively. It can be found that after the installation of the MSB sound barrier, the sound pressure curve at R2 is obviously much smaller than that at S2 at different moments, and the maximum peak value in the time domain is reduced from 2 Pa to 0.5 Pa, which is more than one order of magnitude lower. In the low-frequency band below 100 Hz, there are fluctuations in the frequency-domain sound pressure amplitude at point R2, and the decrease in the sound pressure amplitude is small. At frequencies above 100 Hz, the sound pressure curve at point R2 begins to show a decreasing trend. Compared with point S2, the frequency domain sound pressure amplitude and average value at point R2 decreased significantly. The above results show that the installation of the MSB sound barrier can effectively isolate the noise propagation from the source point to the receiving point.

In order to further quantitatively analyze the insertion loss of the MSB, the calculation of the relevant parameters is carried out in combination with equation (8). The summarized results are shown in Table 1.

Before the installation of the sound barrier, the sound pressure level of S1 is 68.1,the sound pressure level of S1 is 63.2, and the sound pressure level difference between the two is 1.9 dB. D1 is mainly generated by the acoustic friction dissipation of the air or the acoustic absorption of the road surface in the process of noise propagation. After the installation of the sound barrier, the sound pressure levels of S2 and R2 are 76.2 dB and 67.0 dB, respectively. The difference in sound pressure level between the two, D2, is 9.2

Table 1
Summary of results (in dB).

Test No.	Source points	Reception points	difference	insertion loss
1	68.1	66.2	1.9	7.3
2	76.2	67.0	9.2	

dB. However, it is worth noting that D2 includes the noise reduction contribution of D1 in addition to the noise reduction contribution of the MSB sound barrier. Therefore, the actual insertion loss of MSBs is the difference between D2 and D1, i.e., 7.3 dB. In summary, after installing 2m-high and 200m-long MSBs in highway traffic or buildings projects, the noise reduction effect can be effectively improved by 7.3 dB.

6. Conclusion and discussion

An acoustic metamaterial sound barrier (MSB) based on a genetic algorithm (GA) has been optimally designed for efficient noise reduction of traffic noise. Simulation results show that the absorption coefficient of the MSB is more than 0.8 in the range of 700 Hz-8 kHz. Compared with the existing sound barriers, the absorption coefficient of MSB is significantly improved in the low and medium frequency range. MSB has good sound absorption for sound waves with different incidence angles. It is proved that the impedance matching effect is the key to realizing near-perfect sound absorption, and the acoustic transmission loss curves of MSB show that the sound insulation at frequencies above 200 Hz reaches more than 35.0 dB. The experimental results of sound absorption coefficient (SAC) and sound transmission loss (STL) are consistent with the simulation results. After 400Hz, the SAC of both reaches above 0.6, and the absorption coefficients of both of them are stabilized at about 0.85 at the frequency above 800Hz. At frequencies above 200Hz, the STL values are above 30 dB. At frequencies above 800Hz, the experimentally measured STL values are all above 50 dB. Due to the nonlinearity of the material, the sealing of the device, and other interference factors, the two in the peak value of the existence of certain differences The calculation of insertion loss of MSBs in the field was evaluated by the test method of the relevant specifications. The results show that the installation of MSBs reduces the propagation of traffic or buildings noise by 7.3 dB order of magnitude.

The acoustic metamaterials-genetic algorithm collaborative optimization design method and MSBs structure proposed in this study are expected to show unique advantages in realizing the adaptive design of noise-reducing structures and developing low-frequency broadband noise-reducing systems. Subsequent research can rely on this methodology system to expand to the cross-fertilization design of gradient cavity structures, underwater acoustic absorbing metamaterials, acoustic black holes and multifunctional composite metamaterials. Meanwhile, the design efficiency of acoustic metamaterial structures and their comprehensive acoustic/mechanical performance can be further improved by introducing deep learning algorithms for optimization.

CRediT authorship contribution statement

Xinhao Zhang: Software, Methodology, Funding acquisition, Formal analysis. **Caiyou Zhao:** Formal analysis, Data curation. **Duojia Shi:** Writing – original draft. **Pengzhan Liu:** Visualization, Resources. **Tao Lu:** Writing – original draft, Funding acquisition. **Rong Chen:** Validation, Project administration. **Ping Wang:** Writing – review & editing, Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

The scientific research project of China Academy of Railway Sciences Co., Ltd. (No. 2024YJ176); The Fundamental Research Funds for the Central Universities (2682025GH008).

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jobe.2025.113090.

Data availability

Data will be made available on request.

References

[1] K. Sivakumaran, J.A. Ritonja, H. Waseem, L. AlShenaibar, E. Morgan, S.A. Ahmadi, A. Denning, D.S. Michaud, R.L. Morgan, Impact of noise exposure on risk of developing stress-related health effects related to the cardiovascular system: a systematic review and meta-analysis, Noise Health 24 (114) (2022) 107.
[2] H.M. Lee, Z.M. Wang, K.M. Lim, et al., Investigation of the effects of sample size on sound absorption performance of noise barrier, Appl. Acoust. 157 (2020) 106989.
[3] Toshiki Kitagawa, Noise and aerodynamics research on the effect of railway noise barrier, RTR/Report 14 (9) (2000) 11–16.
[4] Byung-Joo Jin, Hyun-Sil Kim, Hyun-Ju Kang, Jae-Seung Kim, Sound diffraction by a partially inclined noise barrier, Appl. Acoust. 62 (9) (2001) 1107–1121.
[5] M. Lv, Q. Li, Z. Ning, et al., Study on the aerodynamic load characteristic of noise reduction barrier on high-speed railway, J. Wind Eng. Ind. Aerod. 176 (2018) 254–262.
[6] V. Laxmi, C. Thakre, R. Vijay, Evaluation of sound barriers based on geometries and materials: a review, Environ. Sci. Pollut. Res. 29 (2022) 1729–1745, https://doi.org/10.1007/s11356-021-16944-2.
[7] A. Materials, Acoustic and thermal insulating materials, Building Decora. Mater. (2011) 359–374, https://doi.org/10.1533/9780857092588.359.
[8] G. Ma, P. Sheng, Acoustic metamaterials: from local resonances to broad horizons, Sci. Adv. 2 (2016) el501595.
[9] S.A. Cummer, J. Christensen, A. Alù, Controlling sound with acoustic metamaterials, Nat. Rev. Mater. 1 (2016) 16001.
[10] H. Ge, M. Yang, C. Ma, et al., Breaking the barriers: advances in acoustic functional materials, Natl. Sci. Rev. 5 (2018) 159–182.
[11] R. Martinez-Sala, J. Sancho, J.V. Sánchez, et al., Sound attenuation by sculpture, Nature 378 (1995) 241.
[12] V.M. Garcia-Chocano, S. Cabrera, J. Sánchez-Dehesa, Broadband sound absorption by lattices of microperforated cylindrical shells, Appl. Phys. Lett. 101 (2012) 184101.
[13] S.A. Cummer, J. Christensen, A. Alù, Controlling sound with acoustic metamaterials, Nat. Rev. Mater. 1 (2016) 16001.
[14] H. Ge, M. Yang, C. Ma, et al., Breaking the barriers: advances in acoustic functional materials, Natl. Sci. Rev. 5 (2018) 159–182.
[15] B. Assouar, B. Liang, Y. Wu, et al., Acoustic metasurfaces, Nat. Rev. Mater. 3 (2018) 460–472.
[16] N. Fang, D. Xi, J. Xu, et al., Ultrasonic metamaterials with negative modulus, Nat. Mater. 5 (2006) 452–456.
[17] S.H. Kim, S.H. Lee, Air transparent soundproof window, AIP Adv. 4 (2014) 117123.
[18] A. Melnikov, M. Maeder, N. Friedrich, et al., Acoustic metamaterial capsule for reduction of stage machinery noise, J. Acoust. Soc. Am. 147 (2020) 1491–1503.
[19] Y. Cheng, C. Zhou, B.G. Yuan, et al., Ultra-sparse metasurface for high reflection of low-frequency sound based on artificial Mie resonances, Nat. Mater. 14 (2015) 1013–1019.
[20] J. Yang, J.S. Lee, H.R. Lee, et al., Slow-wave metamaterial open panels for efficient reduction of low-frequency sound transmission, Appl. Phys. Lett. 112 (2018) 091901.
[21] C. Liu, J. Shi, W. Zhao, et al., Three-dimensional soundproof acoustic metal cage, Phys. Rev. Lett. 127 (2021) 084301.
[22] A. Merkel, G. Theocharis, O. Richoux, et al., Control of acoustic absorption in one-dimensional scattering by resonant scatterers, Appl. Phys. Lett. 107 (2015) 244102.
[23] X. Wu, K.Y. Au-Yeung, X. Li, et al., High-efficiency ventilated metamaterial absorber at low frequency, Appl. Phys. Lett. 112 (2018) 103505.
[24] T. Lee, T. Nomura, E.M. Dede, et al., Ultrasparse acoustic absorbers enable fluid flow and visible-light controls, Phys. Rev. Appl. 11 (2019) 024022.
[25] X. Su, D. Banerjee, Extraordinary sound isolation using an ultrasparse array of degenerate anisotropic scatterers, Phys. Rev. Appl. 13 (2020) 064047.
[26] S. Kumar, T.B. Xiang, H.P. Lee, Ventilated acoustic metamaterial window panels for simultaneous noise shielding and air circulation, Appl. Acoust. 159 (2020) 107088.
[27] S. Kumar, H.P. Lee, Labyrinthine acoustic metastructures enabling broadband sound absorption and ventilation, Appl. Phys. Lett. 116 (2020) 134103l.
[28] J. Du, Y. Luo, X. Zhao, et al., Bilayer ventilated labyrinthine metasurfaces with high sound absorption and tunable bandwidth, Sci. Rep. 11 (2021) 5829.
[29] G. Ma, M. Yang, Z. Yang, et al., Low-frequency narrow-band acoustic filter with large orifice, Appl. Phys. Lett. 103 (2013) 011903.
[30] H. Zhang, Y. Zhu, B. Liang, et al., Omnidirectional ventilated acoustic barrier, Appl. Phys. Lett. 111 (2017) 203502.
[31] R. Ghaffarivardavagh, J. Nikolajczyk, S. Anderson, et al., Ultra-open acoustic metamaterial silencer based on Fano-like interference, Phys. Rev. B 99 (2019) 024302.
[32] X. Wang, X. Luo, B. Yang, et al., Ultrathin and durable open metamaterials for simultaneous ventilation and sound reduction, Appl. Phys. Lett. 115 (2019) 171902.
[33] M. Sun, X. Fang, D. Mao, et al., Broadband acoustic ventilation barriers, Phys. Rev. Appl. 13 (2020) 044028.
[34] J. Shi, C. Liu, X. Liu, et al., Ventilative meta-window with broadband low-frequency acoustic insulation, J. Appl. Phys. 129 (2021) 094901.
[35] X. Bai, Z. Xiao, H. Shi, K. Zhang, Z. Luo, Y. Wu, Omnidirectional sound wave absorption based on the multi-oriented acoustic meta-materials, Appl. Acoust. 228 (2025) 110344.
[36] J. Deng, N. Gao, Broadband vibroacoustic reduction for a circular beam coupled with a curved acoustic black hole via nullspace method, Int. J. Mech. Sci. 233 (2022) 107641.
[37] P. Firoozy, S. Ebrahimi-Nejad, Broadband energy harvesting from time-delayed nonlinear oscillations of magnetic levitation, J. Intell. Mater. Syst. Struct. 31 (5) (2020) 737–755.
[38] M. Ghoudjani, M. Ravanbod, S. Ebrahimi-Nejad, Genetic-algorithm-assisted design of chiral honeycomb membrane acoustic metamaterials for broadband noise suppression, J. Intell. Mater. Syst. Struct. 35 (18) (2024) 1426–1443.
[39] N. Gao, H. Yu, J. Liu, et al., Experimental investigation of composite metamaterial for underwater sound absorption, Appl. Acoust. 211 (2023) 109466.
[40] N. Gao, Z. Zhang, X. Liang, et al., Broadband multi-unit composite metamaterial for simultaneous sound wave and electromagnetic wave absorption, Mater. Des. 251 (2025) 113671.
[41] M. Ghorbany, S. Ebrahimi-Nejad, M. Mollajafari, Global-guidance chaotic multi-objective particle swarm optimization method for pneumatic suspension handling and ride quality enhancement on the basis of a thermodynamic model of a full vehicle. Proceedings of the IMechE, Part D, J. Automobile Eng. 237 (14) (2023) 3334–3352.
[42] X. Zhang, Q. Yu, C. Zhao, et al., Modular reverse design of acoustic metamaterial and sound barrier engineering applications: high ventilation and broadband sound insulation, Thin-Walled Struct. 196 (2024) 111498.
[43] Lige Chang, Xiaowen Li, Zengrong Guo, Yajun Cao, Yuyang Lu, Rinaldo Garziera, Hanqing Jiang, On-demand tunable metamaterials design for noise attenuation with machine learning, Mater. Des. 238 (2024) 112685.
[44] L. Wu, Z. Zhai, X. Zhao, et al., Modular design for acoustic metamaterials: low-frequency noise attenuation, Adv. Funct. Mater. 32 (13) (2021) 2105712.
[45] Bowei Fang, Rui Zhang, Tiannin Chen, Wenfei Wang, Jian Zhu, Wei Cheng, Bandgap optimization and inverse design of labyrinth metamaterials for sound insulation, J. Build. Eng. 86 (2024) 108898.
[46] K. Deb, H.G. Beyer, Self-adaptive genetic algorithms with simulated binary crossover, Evol. Comput. 9 (2) (2001) 137–221.
[47] Nansha Gao, Jing Liu, Jie Deng, et al., Design and performance of ultra-broadband composite meta-absorber in the 200Hz-20kHz range, J. Sound Vib. 574 (2024) 118229.
[48] Yi Fang, Xin Zhang, Jie Zhou, Acoustic porous metasurface for excellent sound absorption based on wave manipulation, J. Sound Vib. 434 (2018) 273–283.
[49] N. Gao, J. Wu, K. Lu, et al., Hybrid composite meta-porous structure for improving and broadening sound absorption, Mech. Syst. Signal Process. 154 (2021) 107504.
[50] V. Gupta, A. Singh, B. Bhattacharya, Wave propagation study in metamaterial sandwich structure with periodically inserted hourglass resonators, Proc. SPIE 12483 (000) (2023) 9.
[51] T. Mikami, S. Kitamura, et al., Brillouin-Wigner theory for high-frequency expansion in periodically driven systems: application to Floquet topological insulators, Phys. Rev. B. 93 (2016) 144307.